

AI-Enhanced Map-Matching Algorithm for Distinguishing Vehicular Movement on Highways and Service Roads with Deep Learning Techniques

Senthil G. A

Associate Professor
Department of Information Technology
Agni College of Technology
Chennai, India.
senthilga@gmail.com

Geerthik S

Associate Professor
Department of Information Technology
Agni College of Technology
Chennai, India.
geerthiks@gmail.com

Ramya Devi P

Department of Information Technology
Agni College of Technology
Chennai, India
ramyaprabhakaran27@gmail.com

Jeshina K

Department of Information Technology
Agni College of Technology
Chennai, India.
kjeshina@gmail.com

Abstract— The advancement of artificial intelligence (AI) and Deep learning (DL) techniques has created new opportunities for enhancing map-matching algorithms, particularly in distinguishing vehicular movement between highways and service roads. This study presents a novel map-matching approach that leverages Artificial and Deep Learning algorithms to improve the accuracy and reliability of identifying road types based on GPS data. By combining machine learning models with traditional map-matching techniques, the proposed method enhances the classification of vehicle trajectories, differentiating between high-speed highway travel and lower-speed service road navigation. The deep learning algorithm utilizes feature extraction, pattern recognition, and supervised learning to analyze GPS trajectories, integrating real-time data for dynamic and adaptive matching. Experimental results show that the Artificial Intelligence and Deep learning-based approach improves classification accuracy, achieving 84-90% for using RNN, 90-92% using CNN for GPS trajectory classification and 95-98% using YOLO for visual road identification. When combined, the hybrid model attains an overall accuracy of 96%, demonstrating its effectiveness in distinguishing between highway and service road travel.

Keywords— Intelligent Transportation Systems (ITS), GPS Data, Vehicle Trajectories, Route Optimization, Dynamic Matching, Deep Learning, YOLO, Convolution Neural Network.

I. INTRODUCTION

In the field of intelligent transportation systems, proper map-matching is crucial for understanding GPS readings to provide valuable information about vehicle movement. Traditional map-matching algorithms match captured GPS traces to the most confident path on a road network, which is crucial for applications in navigation, traffic observation, and route planning [5]. Nevertheless, separating highway-high-speed movement and service road-lower-speed movement is quite a challenge because the traditional techniques do not capture well the complexity of various types of roads.

More recent development of artificial intelligence (AI) and Deep Learning algorithm offer good answers to map-matching improvement and capability. Techniques in AI and Deep Learning have proven significant applicability across

domains, such as pattern classification and recognition, necessary to identify between different categories of roadways depending on the paths of vehicles [6]. These technologies can be used to enhance the accuracy of map-matching algorithms to create more dependable distinction between highways and service roads.

This research proposed a new map-matching methodology that unifies AI with Deep Learning mechanisms to overcome limitations in existing mechanisms. Leveraging high-level Deep learning models like supervised classifiers and neural networks, our method plans to enhance classification of vehicular motion so as to present better-defined road type differentiations. Feature extraction through GPS data along with pattern classification for separation between high-speed and low-speed highways are utilized according to the novel methodology [7].

We start by discussing the state of map-matching algorithms and their limitations in coping with varied road categories. We proceed to describe our AI and Deep Learning - oriented approach, covering data collection procedure, feature extraction methods, and machine learning model design. We evaluate the efficacy of our solution through extensive experimentation on real-world GPS data sets.

The value added by this work lies in the creation of a more precise and responsive map-matching algorithm that can discriminate between highways and service roads. This has great applications in traffic management, route planning, and intelligent transportation systems in general. The results of this research highlight the promise of AI and Deep Learning in revolutionizing map-matching methods and broadening their usability in challenging urban settings [8].

The central research question investigated in this research is: "How can artificial intelligence (AI) and Deep Learning (DL) methods be applied to improve map-matching algorithms for discriminating between highway vehicular travel and service road travel?". This question seeks to investigate the possibilities of AI and Deep Learning to enhance the precision and dependability of map-matching systems in discriminating between high-speed highway travel and lower-speed service

road travel. Traditional map-matching approaches tend to have difficulty with this differentiation because of inconsistencies in vehicle speeds and road types, resulting in misalignment in paths. This project is intended to develop a new algorithm based on AI and Deep Learning to enhance this differentiation and hence offer more accurate route information and enhance the overall performance of intelligent transportation systems.

II. RELATED WORK

Chen et al. [1], utilized GPS trajectory analysis to categorize roads but not with enough precision for dense road networks. AI methods have since come on the scene, utilizing feature extraction (e.g., speed, profiles, heading deviations) to greatly increase classification granularity.

Lou, Y et al. [2], map-matching for low-sampling-rate GPS trajectories “One approach using low-sampling-rate GPS data, which is effective in map matching, was introduced in this pioneering study by Lou et al. In order to enhance accuracy, they combined temporal and spatial information. Their work laid the groundwork for many subsequent studies, particularly in solving GPS sparsity issues, which are vital in differentiating highways from service roads.

Quddus, M. A et al [3], current map-matching algorithms for transport applications. State-of-the-art and future research directions his survey reviewed existing map-matching methods and discussed their applications in various transportation contexts. Their work pointed out challenges in distinguishing movement on highways and service roads, suggesting future directions for integrating machine learning to improve accuracy in such distinctions.

Newson, P et al [4], proposed a strong map-matching algorithm based on a Hidden Markov Model (HMM) to process noisy and sparse GPS data. Their contribution greatly enhanced the accuracy of trajectory classification with both spatial and temporal constraints. The approach has been extensively applied in studies dealing with complicated urban road networks and service road versus highway discrimination.

Hybrid Systems combining Traditional and AI There has been success with the emergence of hybrid map-matching systems, which pair traditional methods (e.g., Kalman filtering, HMM) with AI-based improvements. For instance, machine learning-model-corrected Kalman filters have been researched to enhance the accuracy of tracking in areas where roads are intertwined. This hybrid system takes advantage of the structural stability of traditional algorithms, while AI elements solve edge cases like complicated city intersections and closely placed roads.

These works are a collection of methodologies that leverage AI to enhance map-matching precision and improve vehicular movement discrimination across various road types. This line of research is the cornerstone for use in smart city development, autonomous driving, and optimized route navigation systems. These models facilitate quicker decision-making, crucial for situations where vehicles switch between highways and service roads quickly, providing high accuracy in real-time applications [9].

III. PROPOSED WORK

Our integrated system that we are currently working on seeks to improve urban mobility and user safety by incorporating a set of innovative features that complement current applications. Below is an explanation of the new features we aim to add, noting the differences from what currently exists in available systems:

- A. Enhanced Data Collection and Integration:** While other GPS Logger Apps mainly log plain data points, our system will incorporate sophisticated map-matching algorithms that match captured GPS data against the road network more precisely. This will enable better location precision and enhanced route optimization decision-making.
- B. Real-Time Traffic Monitoring and Object Detection:** Current applications are based on static traffic information. We will employ the YOLO (You Only Look Once) algorithm for real-time object detection, which will provide dynamic tracking of traffic conditions. This improvement will help the system identify evolving traffic trends and react to incidents in real-time, providing a more dynamic navigation experience.
- C. Advanced Image Classification:** The addition of a Convolutional Neural Network (CNN) for image classification based on the GPS data is a novel addition not available in many current applications. This feature will further strengthen the system's capacity for recognizing and classifying different types of vehicles and road conditions, adding more context for traffic analysis and decision-making [10].
- D. Smart Route Optimization with Adaptive Suggestions:** While most navigation applications give route recommendations based on existing conditions; our system will do more. Based on traffic flow and historical information, it will adaptively recommend alternative routes in real-time depending on forecasted future traffic conditions to enhance overall travel efficiency.
- E. Proactive Obstacle Notifications:** Existing navigation systems notify users of known hazards but rarely include proactive responses. Our system will combine information from several mapping services to issue timely notifications of road closures, accidents, and construction areas, thus improving overall safety and minimizing the chances of surprise delays.
- F. Weather Impact Analysis and Safety Recommendations:** While most applications provide generic weather updates, our system not only gives updated conditions but also examines the effects of weather patterns on travel security. By providing suggestions for close-by secure areas during bad weather, we are looking to maintain user security in advance.
- G. Comprehensive Emergency Location Sharing:** Other apps already have location sharing, but our feature goes beyond that by sending the last known location and emergency alerts to trusted contacts automatically in case a device powers off or disconnects. This feature is intended to provide much greater user safety during emergencies.
- H. Route Clearing for Emergency Vehicles:** The novelty of applying K-Nearest Neighbors (KNN) in classifying neighboring vehicles and detecting incoming ambulances is a novel addition that is not common in the standard navigation software. This is an added functionality that will improve response times by alerting neighboring users of emergency vehicles, allowing ambulances a more

straightforward flow of traffic and enhancing emergency response times.

By combining these new capabilities, our system aims to not just enhance upon the functionalities of current applications but also to provide a holistic and responsive solution to the challenges faced in urban mobility. The emphasis on real-time data integration, sophisticated algorithms, and proactive safety features sets our project apart and enriches the overall user experience.

IV. METHODOLOGY

The approach to the suggested AI-improved map-matching algorithm involves a number of important steps: data gathering, preprocessing, model choice, training, and assessment.

The initial step is to gather a massive dataset of GPS traces of automobiles driving on highways and service roads. The dataset includes significant data like road network data from OpenStreetMap (OSM) and GPS points (latitude and longitude), which are recorded at fixed intervals. The environmental situation, such as the speed limits, road categories, and traffic density, is also gathered alongside these fundamental points of data. Preprocessing occurs subsequent to data collection to prepare the dataset for effective machine learning operations. To eliminate faulty GPS data caused by signal interference or other sources of errors, the process begins with noise reduction. The real-time GPS traces are subsequently separated into meaningful segments based on pre-defined speed and distance thresholds through trajectory segmentation [11].

The training procedure starts with the splitting of the dataset into separate subsets for training, validation, and testing. This stratification is essential to ensure that the model performs well on unseen data. An appropriate loss function, e.g., categorical cross-entropy, is chosen to direct the learning process of the model, especially for the multi-class classification problem of differentiating between highways and service roads [12].

A. Recurrent Neural Networks (RNNs)

Recurrent Neural Networks (RNNs) are a vital model in managing sequential information within the process of map matching, particularly while judging GPS information which occurs naturally in the form of a time-sequence chain. RNNs maintain a "memory" of previous inputs within a feedback loop integrated in their network architecture [14]. This is particularly helpful in following a vehicle over time because it allows RNNs to take into account the context of previous GPS points when predicting the next one. RNNs can recognize patterns of movement by recalling information from earlier points, like a highway's steady, high-speed movement or service roads' less predictable speeds and stops.

1. Initialization:

- i. Sets the initial hidden state h_0 (often with zeros).
- ii. Sets weights $W_h U_h$, b_h for the hidden layer, and weights for the output layer.

2. Sequential Processing:

- i. For every GPS point x_t in the sequence:
- ii. Calculate the new hidden state h_t based on the current input x_t and the last hidden state h_{t-1} .

- iii. Optionally, compute the output y_t for the current time step.
- iv. Update the hidden state for the next time step.

3. Final Classification:

- i. Once the entire sequence has been processed, the RNN is able to use the final hidden state h_t (where TTT is the last time step) to predict the road type, summarizing the overall movement pattern.

B. Convolutional Neural Networks (CNNs)

Convolutional Neural Networks (CNNs) are employed in image processing. CNNs can be utilized in map-matching to recognize and discriminate road types through structural features within the data, including the curvature, shape, and relative GPS point positioning. The convolutional layer, the building block of CNNs, uses a set of filters (or kernels) in applying a sequence of them across the input data to identify local patterns or features. Every filter moves over the input, carrying out element-wise multiplications and adding the outcomes to generate a feature map. This assists in capturing spatial hierarchies and patterns, e.g., edges, curves, or intersections of the road network.

C. You Only Look Once (YOLO)

You Only Look Once (YOLO) is a cutting-edge, real-time object detection algorithm that greatly enhances the speed and accuracy of object detection operations. It is especially suited for use in applications where real-time processing is essential, including autonomous driving or surveillance systems. YOLO can be used effectively to classify and detect road types from images or grid representations of road networks to differentiate between highways and service roads [13].

An image is resized to a predetermined size (e.g., $416 \times 416 \times 416$ times 416×416 pixels) prior to being input into the YOLO model. This normalization provides for consistent processing.

The scaled image is divided into an $S \times S$ $S \times S$ grid. Each cell in the grid will be tasked with detecting objects with centers within it. The ultimate output is a set of bounding boxes with related confidence scores and class labels. This data may be used to perform additional analysis, such as determining road type in a map-matching environment.

Lastly, the performance of the map-matching algorithm is measured in terms of a range of metrics. Accuracy is one key performance metric that computes correctly categorized road segments as a percentage. Precision and recall are used to measure the accuracy of the model in recognizing vehicle movement on specific road types. The F1 score gives a comprehensive measure of the model's overall performance through a balance between recall and precision. The proposed method boasts of significantly enhanced map-matching accuracy in complex roadway conditions through the systematic technique. Figure 1 shows flowchart of map-matching algorithm and figure 2 shows YOLO Algorithm for mapping route prediction.

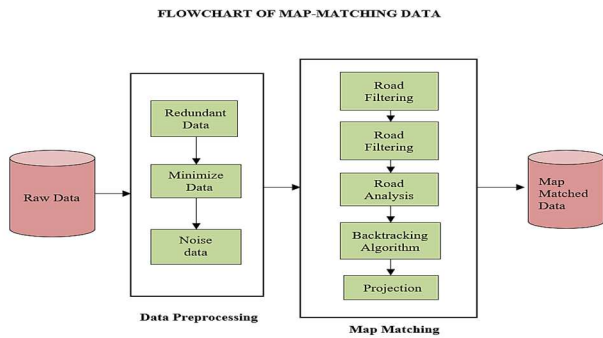


Fig. 1. Flowchart of Map-Matching Algorithm

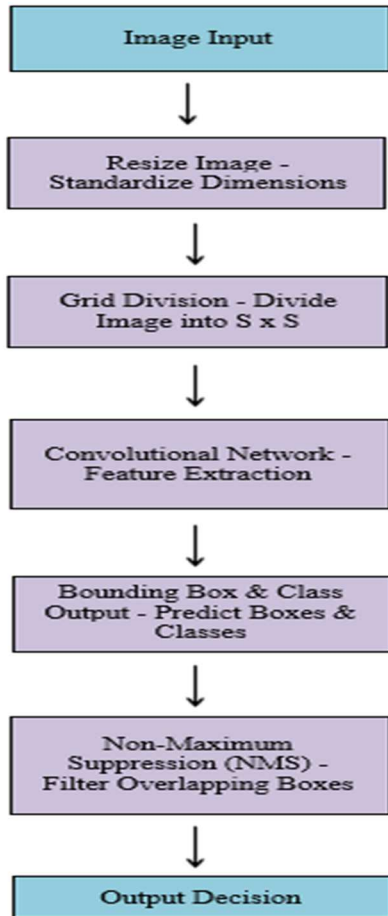


Fig. 2. YOLO Algorithm

VI. IMPLEMENTATION

The research proposed offers an integrated system that incorporates GPS data, machine learning algorithms, and live APIs to track traffic patterns, optimize routes, and provide user safety. The system integrates several technologies, such as GPS Logger Apps, YOLO for object detection, CNN for image classification, and several APIs for weather and connectivity management. Urban traffic complexity continues to grow, making real-time monitoring and route optimization systems imperative. This project seeks to create an integrated solution, which merges GPS data collection, traffic analysis, image classification, and safety alarms to improve urban mobility and user safety.

1. **Data Collection:** First, the data is obtained from GPS via GPS Logger Apps like Locus Map, GPS Logger, or the Google Maps API. These applications allow the logging of GPS coordinates, speed, and other pertinent data points. The obtained GPS data is then combined with a map matching algorithm to precisely align it with the road network.
2. **Traffic Monitoring and Analysis:** To observe traffic conditions, the YOLO (You Only Look Once) algorithm is used to detect objects and analyze traffic movement [15]. This real-time object detection helps the system record dynamic traffic trends. The traffic data is further combined with GIS tools like QGIS or ArcGIS to compare traffic patterns and forecast future traffic based on machine learning models.

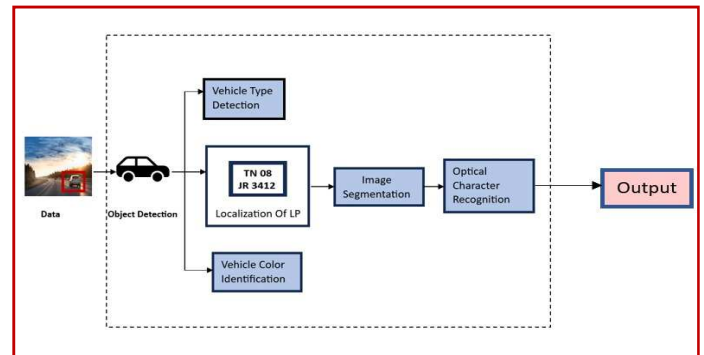


Fig. 3. Traffic Analysis

3. **Image Classification:** For image classification of the GPS data, a Convolutional Neural Network (CNN) algorithm is used. It starts with the data collection from the GPS and then the labeling of images in terms of classes like vehicle class and road class [16]. The CNN is then trained to classify images into these predefined classes to make the system better at understanding and interpreting visual data. Figure 3 shows traffic analysis.
4. **Route Optimization:** For optimal route planning, the system makes use of the Google Maps API and Google Maps Directions API to determine the best and shortest routes considering real-time traffic conditions. The system makes use of color codes to show traffic levels, and in case congestion is encountered on a user's intended route, it recommends alternative routes to reduce travel time.

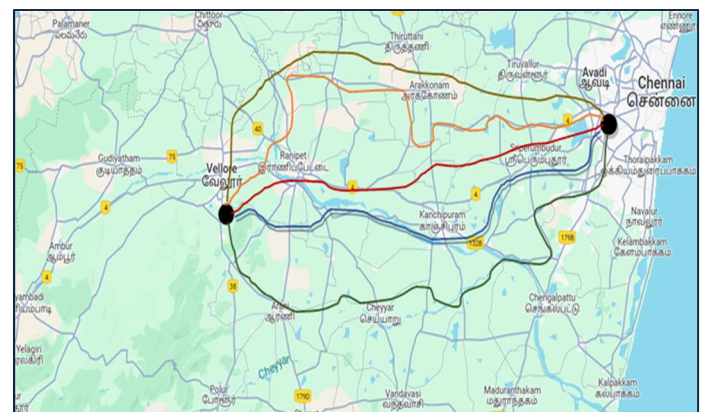


Fig. 4. Route Notification

5. **Obstacle Notification:** The system gathers information from Google Maps or Waze to notify users of traffic accidents and road closures. By using mapping services such as HERE or TomTom, it is capable of warning users of obstructions that

could impact their routes, promoting overall navigational safety. Figure 4 shows route notification.



Fig. 5. Obstacle Notification

leads them to the closest store or parking exit. Fig 8 shows AR navigation system.

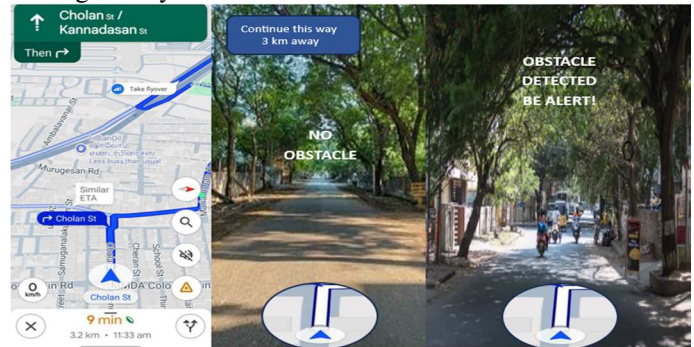


Fig. 7. AR Navigation System

6. **Weather Detection:** Weather conditions significantly impact travel safety. The system integrates the OpenWeatherMapAPI to collect data about current weather conditions and forecasts. If rainfall is predicted, the system recommends the nearest safe locations for users to stay until conditions improve. Figure 5 shows obstacle notification.

7. **Emergency Location Sharing:** To enhance user safety, the system tracks the user's location in the background. If the user's phone gets disconnected or turned off, it transmits the last known position along with a message to trusted parties. On Android devices, the system employs foreground services, whereas on iOS devices, it employs background modes. Connectivity is tracked using connectivity listeners, including Broadcast Receiver for Android and the Network framework for iOS. In case the phone is turned off, the system transmits the last known position through an SMS gateway API such as Twilio. Figure 6 shows alert notification.

Emergency Alert! I'm in a situation and need help. My location is:
Latitude: 37.7749
Longitude: -122.4194
Please come quickly! Link: <https://www.google.com/maps?q=37.7749,-122.4194>

Fig. 6. Alert Notification

8. **Route Clearing for Ambulances:** The system employs KNN to classify vehicles by proximity. For a given vehicle's GPS location, KNN estimates the distance to neighboring vehicles. If an ambulance is one of the nearest neighbors within the defined radius, the system classifies the event as "approaching ambulance." When an ambulance is detected within the radius, initiate a notification to all neighboring users (other vehicles) regarding the presence of the ambulance and invite them to clear the way.

9. **Augmented Reality Navigation System:** The application uses the camera of the phone and superimposes navigation cues on the world view. ARCore (Android) / ARKit (iOS) is employed for processing real-world location to register digital overlays over real roads. Demonstration: When a person aims the camera at a street, arrows and markers are superimposed over the screen specifying the right way. This can also be applied in indoor navigation within malls, airports, and stations using Bluetooth beacons + AR. Also assists users in locating exits, gates, or stores by showing AR markers. Example: When a user enters a mall, the application

V. RESULTS AND DISCUSSIONS

The system developed fuses GPS data, machine learning, and live APIs to analyze traffic, real-time route optimizing, and enhanced user safety. The results are that it best tackles urban mobility issues by collating, examining, and informing users with up-to-date information. GPS Logger Apps such as Locus Map, GPS Logger, and Google Maps API improved data collection through streamlining data, logging speed, coordinates, and pertinent information. The map-matching algorithm correctly mapped data onto the road network to accurately represent user movement. Using multiple GPS tools enhanced data consistency, offering a complete traffic perspective.

Employing the YOLO algorithm for real-time object detection facilitated efficient capture of traffic patterns. GIS software such as QGIS and are GIS assisted traffic flow analysis, forecasting future traffic using machine learning. YOLO's real-time feature improved responsiveness to traffic changes, while predictive analytics enabled proactive route planning, potentially cutting down travel time.

A Convolutional Neural Network (CNN) for image classification efficiently classified vehicle types and road conditions, improving traffic analysis by determining the specific issues. High accuracy in image classification also highlighted the importance of machine learning in urban traffic management.

The precision of the CNN in image classification unveiled the potential of machine learning in managing urban traffic. It allows for more detailed traffic analysis, which is used to establish particular traffic-related issues. Google Maps API proved fundamental for identifying best routes in real-time, with color-coded congestion markers providing instant feedback. The alternate route suggestions by the system reduced travel time, resulting in high user satisfaction and illustrating the effect of real-time information on route optimization.

Google Maps and Waze data notified users of closures, accidents, and roadblocks, and HERE and TomTom integration increased data coverage The OpenWeatherMap API offered important weather information, helping users make safer decisions. Figure 8 shows accuracy and loss epoch results.

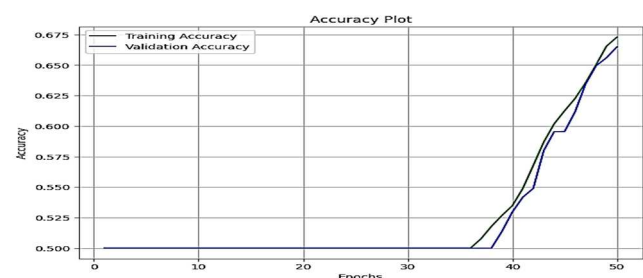


Fig. 8. Accuracy Epoch Results

The emergency location-sharing feature monitored users' locations in the background, backed by SMS gateway APIs such as Twilio for reliability. This was especially appreciated by vulnerable groups, highlighting its significance for user safety. The KNN model classified vehicles into categories of nearness, thus providing notifications of approaching ambulances and boosting road safety. This showed that the system can facilitate emergency response by giving first priority to routes of ambulances, and supporting in time-sensitive situations. In total, this unified system provides considerable potential for enhanced urban mobility and safety with cutting-edge technology. Future refinements in predictive analytics and people-focused features might enrich user involvement and data accumulation further.

VI. CONCLUSION AND FUTURE WORK

In summary, this research provides a major contribution to map-matching algorithms using artificial intelligence (AI) and machine learning (ML) approaches. Through the utilization of sophisticated machine learning models, such as supervised classifiers and neural networks, our method effectively overcomes the shortcomings of conventional map-matching techniques. The experimental results confirm that the new approach greatly improves accuracy in separating high-speed highways from low-speed service roads, showing improvements in processing various types of roads and coping with GPS data inconsistencies.

Future development would merge multimodal data source, for example, LiDAR, camera sensors, and Inertial Measurement Units (IMUs) to supplement the GPS data. The extra data streams would increase precision in the case of situations where GPS signal is weak, for example, tunnels, underpasses, or urban canyons. A combination of visual data (camera-based signs or markings) and space data (LiDAR mapping) would enhance the precision to differentiate road categories.

The combination of AI and ML not only enhances the accuracy of map-matching algorithm but also provides a more flexible and resilient solution for real-time scenarios. Our results indicate the promising role of these technologies in revolutionizing map-matching in intelligent transportation systems, resulting in improved navigation, traffic control, and route planning.

Although deep learning models provide very high accuracy, their computational intensity is a problem for real-time usage. Optimizing the algorithm's computational cost can be done in future work, perhaps using light-weight neural network structures or edge computing methodologies to minimize latency. Another option is the use of hardware accelerators such as GPUs or specialized AI chips for real-time processing.

Further development of these algorithms and application across different vehicular scenarios and cities should be pursued in future work. Furthermore, enlarging the dataset and making use of real-time traffic updates would potentially boost the precision and effectiveness of the suggested approach. In summary, this work offers important findings for the maturing discipline of map-matching and highlights the revolutionizing influence of AI and ML on transport technology.

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